

PROJECT REPORT

CUSTOMER CHURN PREDICTION & RETENTION FRAMEWORK

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Certificate

This is to certify that the project entitled **Customer Churn Prediction & Retention Framework** has been successfully completed by **Boda Dhanesh** in partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Science & Engineering.

Declaration

I hereby declare that this report is my original work and has not been submitted elsewhere for any degree or diploma.

Signature: **Boda Dhanesh**

Acknowledgement

I express my sincere gratitude to my faculty members, project guide, and Ganpat University for their continuous support and guidance throughout this project.

Abstract

Customer churn is a major challenge for businesses because losing customers reduces revenue and increases acquisition costs. This project develops a data-driven framework to identify customers who are likely to discontinue services. The Telco Customer Churn dataset is analyzed using Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-Learn. The project includes data preprocessing, exploratory data analysis, key metric identification, dashboard development, and predictive modeling using Support Vector Machine (SVM).

Contents

1	Introduction	10
1.1	Project Overview	10
1.2	Background of the Study	10
1.3	Problem Statement	10
1.4	Project Objectives	10
1.5	Scope of the Project	11
1.6	Project Deliverables	11
1.7	Report Organization	11
2	Requirement Analysis	12
2.1	Domain Study	12
2.2	Existing System Analysis	12
2.3	Proposed System	12
2.4	Functional Requirements	12
2.5	Non-Functional Requirements	12
2.6	Tools and Technologies Used	13
3	Data Collection and Understanding	14
3.1	Data Source	14
3.2	Data Description	14
3.3	Data Structure	14
3.4	Data Dictionary	14
3.5	Data Quality Assessment	14
4	Data Preprocessing	15
4.1	Data Cleaning	15
4.2	Handling Missing Values	15
4.3	Duplicate Removal	15
4.4	Data Transformation	15
4.5	Data Validation	15
4.6	Final Prepared Dataset	15

5	Data Analysis and Modeling	16
5.1	Exploratory Data Analysis (EDA)	16
5.1.1	Customer Churn Distribution	16
5.1.2	Contract Type Distribution	17
5.1.3	Senior Citizen Distribution	17
5.1.4	Paperless Billing Usage	18
5.1.5	Gender Distribution	18
5.2	Data Relationships	19
5.2.1	Churn Rate by Contract Type	19
5.2.2	Churn Rate by Internet Service	19
5.2.3	Churn Rate by Payment Method	20
5.2.4	Churn Rate by Gender	20
5.2.5	Churn Rate by Customer Tenure	21
5.2.6	Correlation Analysis	21
5.3	Key Metrics Identification	21
5.4	Business Rules and Logic	21
5.5	Data Modeling Approach	22
5.5.1	SVM Overview	22
5.5.2	Why SVM for Churn Prediction	22
6	Dashboard Design and Development	23
6.1	Dashboard Planning	23
6.2	Data Model Design	23
6.3	Dashboard Components	23
6.4	Visualizations Used	23
6.5	Interactive Features	23
6.6	User Interface Design	24
7	Results and Insights	25
7.1	Dashboard Output	25
7.2	Key Findings	25
7.3	Trend Analysis	25
7.4	Comparative Analysis	25
7.5	Generated Insights	25
8	Testing and Validation	26
8.1	Testing Strategy	26
8.2	Test Cases	26
8.3	Validation Process	26
8.4	Performance Evaluation	26

8.5	Limitations	26
9	Deployment	28
9.1	Deployment Process	28
9.2	Publishing and Sharing	28
9.3	User Access and Usage	28
9.4	Documentation	28
10	Conclusion and Future Scope	29
10.1	Project Summary	29
10.2	Objectives Achieved	29
10.3	Learning Outcomes	29
10.4	Future Enhancements	29
A	Dataset Samples	32
B	Dashboard Screenshots	33
C	Formulas and Calculations	34
D	User Guide	35
E	Project Timeline	36

List of Figures

5.1	Customer Churn Distribution	16
5.2	Contract Type Distribution	17
5.3	Senior Citizen Distribution	17
5.4	Paperless Billing Usage	18
5.5	Gender Distribution	18
5.6	Churn Rate by Contract Type	19
5.7	Churn Rate by Internet Service	19
5.8	Churn Rate by Payment Method	20
5.9	Churn Rate by Gender	20
5.10	Churn Rate by Customer Tenure	21
5.11	Correlation Matrix	21

List of Tables

2.1	Tools and Technologies	13
8.1	Test Cases	26

Chapter 1

Introduction

1.1 Project Overview

Customer Churn Prediction & Retention Framework is a data-driven system designed to identify customers who are likely to stop using a company's products or services.

1.2 Background of the Study

Businesses collect large amounts of customer data, but identifying customers who may leave is difficult without analytics and machine learning.

1.3 Problem Statement

Organizations face customer loss due to competition, dissatisfaction, pricing, and service quality issues. The project aims to predict churn and support retention decisions.

1.4 Project Objectives

1. Understand customer churn behavior.
2. Analyze the Telco dataset.
3. Perform data preprocessing.
4. Conduct exploratory data analysis.
5. Identify churn factors.
6. Develop a predictive framework.

1.5 Scope of the Project

The framework can be applied in telecommunications, banking, e-commerce, insurance, subscription services, and online platforms.

1.6 Project Deliverables

- Final project report
- Cleaned dataset
- EDA visualizations
- Dashboard screenshots
- Predictive model results

1.7 Report Organization

This report is organized into ten chapters covering analysis, modeling, dashboard development, testing, deployment, and future scope.

Chapter 2

Requirement Analysis

2.1 Domain Study

Customer churn prediction belongs to the field of data analytics and machine learning.

2.2 Existing System Analysis

Manual customer analysis is slow and often fails to detect hidden churn patterns.

2.3 Proposed System

The proposed system uses data preprocessing, EDA, dashboard analytics, and machine learning to predict churn.

2.4 Functional Requirements

- Load dataset
- Clean data
- Generate visualizations
- Identify churn patterns
- Predict churn

2.5 Non-Functional Requirements

- Accuracy

- Scalability
- Reliability
- Usability

2.6 Tools and Technologies Used

Technology	Purpose
Python	Programming
Google Colab	Development environment
Pandas	Data analysis
NumPy	Numerical computation
Matplotlib	Visualization
Seaborn	Statistical visualization
Scikit-Learn	Machine learning
Power BI	Dashboard development

Table 2.1: Tools and Technologies

Chapter 3

Data Collection and Understanding

3.1 Data Source

The project uses the Telco Customer Churn dataset.

3.2 Data Description

The dataset contains customer demographics, service subscriptions, billing information, and churn status.

3.3 Data Structure

Each row represents a customer and each column represents a feature.

3.4 Data Dictionary

Important attributes include CustomerID, Gender, Tenure, MonthlyCharges, TotalCharges, Contract, PaymentMethod, InternetService, and Churn.

3.5 Data Quality Assessment

Missing values, duplicates, and data types were checked using Pandas functions.

Chapter 4

Data Preprocessing

4.1 Data Cleaning

Incorrect values and inconsistencies were identified and corrected.

4.2 Handling Missing Values

Missing values were checked using:

```
df.isnull().sum()
```

4.3 Duplicate Removal

Duplicate records were removed using:

```
df.drop_duplicates()
```

4.4 Data Transformation

The TotalCharges column was converted to numeric format.

4.5 Data Validation

The cleaned dataset was validated by checking shape, data types, and statistical summaries.

4.6 Final Prepared Dataset

The final dataset was prepared for EDA, dashboarding, and machine learning.

Chapter 5

Data Analysis and Modeling

5.1 Exploratory Data Analysis (EDA)

5.1.1 Customer Churn Distribution



Figure 5.1: Customer Churn Distribution

Observation: Approximately 73.5% of customers remained with the company, while 26.5% churned.

5.1.2 Contract Type Distribution

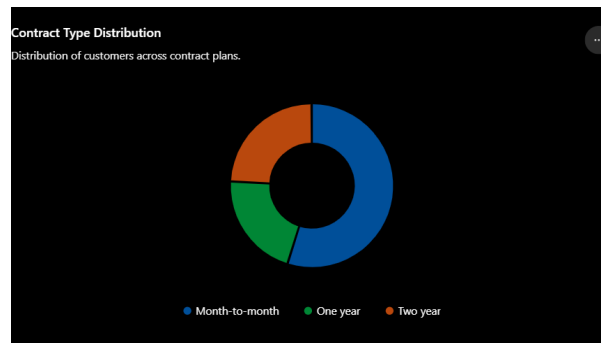


Figure 5.2: Contract Type Distribution

Observation: Most customers prefer Month-to-Month contracts.

5.1.3 Senior Citizen Distribution

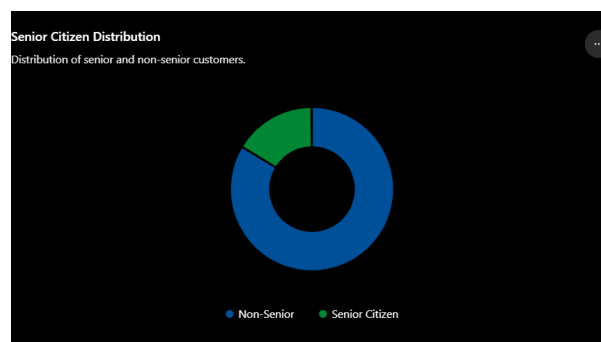


Figure 5.3: Senior Citizen Distribution

5.1.4 Paperless Billing Usage

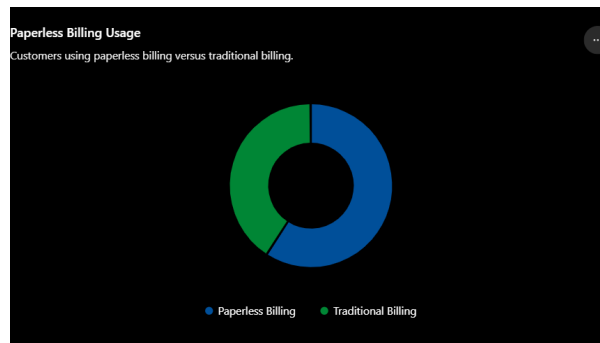


Figure 5.4: Paperless Billing Usage

5.1.5 Gender Distribution

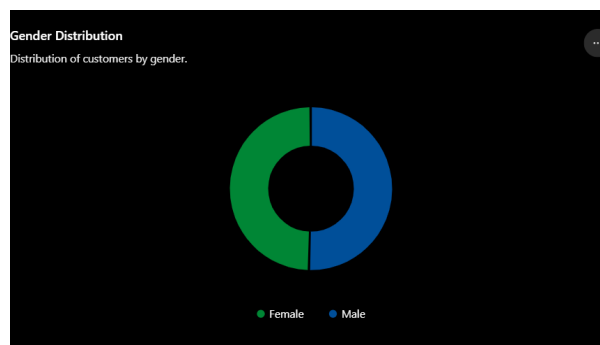


Figure 5.5: Gender Distribution

5.2 Data Relationships

5.2.1 Churn Rate by Contract Type

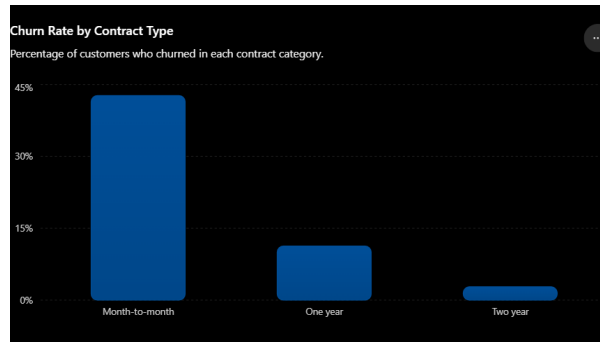


Figure 5.6: Churn Rate by Contract Type

5.2.2 Churn Rate by Internet Service

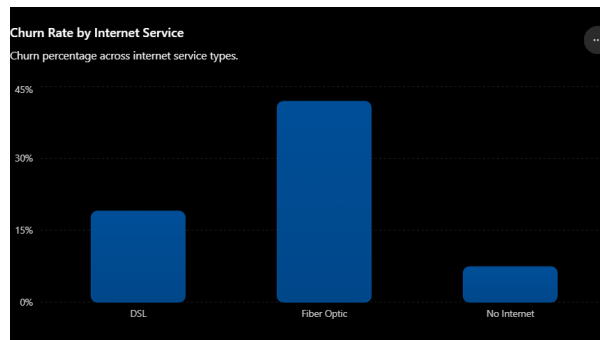


Figure 5.7: Churn Rate by Internet Service

5.2.3 Churn Rate by Payment Method

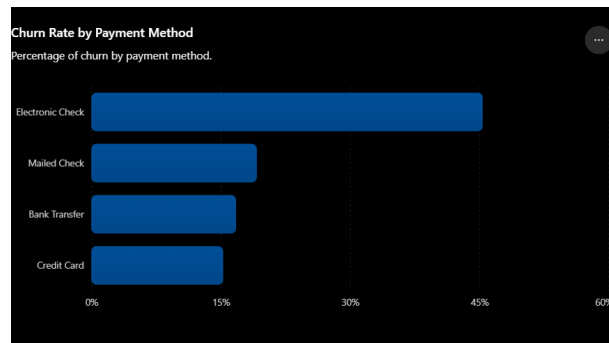


Figure 5.8: Churn Rate by Payment Method

5.2.4 Churn Rate by Gender

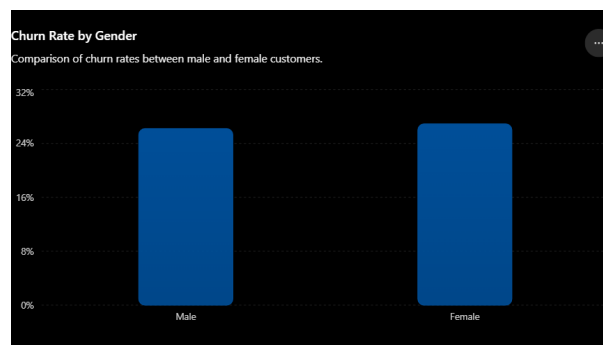


Figure 5.9: Churn Rate by Gender

5.2.5 Churn Rate by Customer Tenure

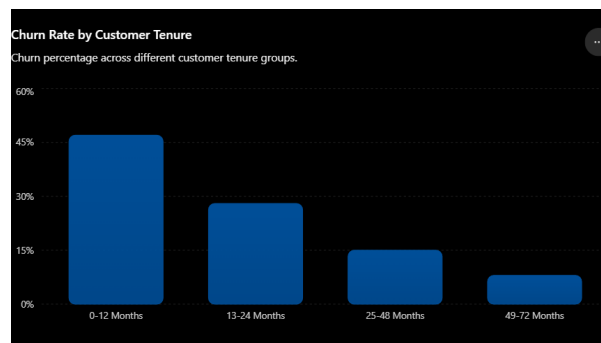


Figure 5.10: Churn Rate by Customer Tenure

5.2.6 Correlation Analysis

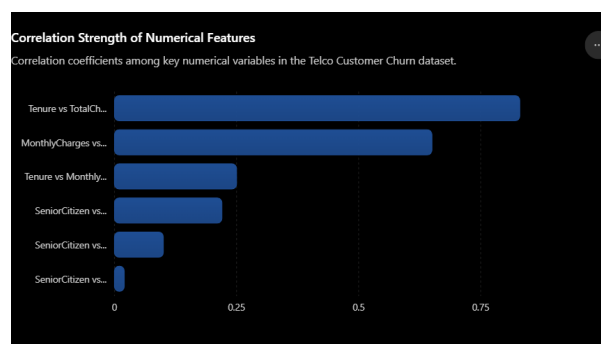


Figure 5.11: Correlation Matrix

5.3 Key Metrics Identification

Important metrics include Churn Rate, Retention Rate, Average Monthly Charges, Average Tenure, Contract-Based Churn Rate, Internet Service Churn Rate, and Payment Method Churn Rate.

5.4 Business Rules and Logic

- Month-to-Month contracts are considered high-risk.
- Tenure less than 12 months indicates higher churn probability.

- Electronic Check users are considered higher risk.
- Fiber Optic customers require additional retention analysis.

5.5 Data Modeling Approach

Support Vector Machine (SVM) was selected for churn classification.

5.5.1 SVM Overview

SVM is a supervised machine learning algorithm used for classification and regression tasks.

5.5.2 Why SVM for Churn Prediction

- High classification accuracy
- Effective for medium-sized datasets
- Works well in high-dimensional spaces
- Suitable for binary classification (Churn/Non-Churn)

Chapter 6

Dashboard Design and Development

6.1 Dashboard Planning

Three Power BI dashboards were planned based on key KPIs.

6.2 Data Model Design

Relationships were created between customer, billing, service, and churn tables.

6.3 Dashboard Components

- KPI cards
- Bar charts
- Pie charts
- Line charts
- Filters and slicers

6.4 Visualizations Used

Visualizations include churn distribution, contract analysis, tenure trends, payment analysis, and service-wise churn.

6.5 Interactive Features

Slicers, drill-downs, cross-filtering, and dynamic KPI updates were implemented.

6.6 User Interface Design

The dashboard was designed with a clean layout, consistent colors, and easy navigation.

Chapter 7

Results and Insights

7.1 Dashboard Output

The Power BI dashboards successfully displayed customer churn patterns and business KPIs.

7.2 Key Findings

- Churn rate is approximately 26.5%.
- Month-to-Month customers show the highest churn.
- Fiber Optic users have elevated churn rates.
- Customers with low tenure are more likely to leave.

7.3 Trend Analysis

Churn decreases as customer tenure increases.

7.4 Comparative Analysis

Long-term contract customers are significantly more stable than short-term contract customers.

7.5 Generated Insights

Retention campaigns should target new customers, Month-to-Month users, and Electronic Check users.

Chapter 8

Testing and Validation

8.1 Testing Strategy

Functional testing, data validation, visualization testing, and model validation were performed.

8.2 Test Cases

Test Case	Expected Result	Status
Load dataset	Dataset loads successfully	Pass
Remove duplicates	No duplicate rows	Pass
Generate charts	Visualizations displayed	Pass
Train SVM model	Model trains successfully	Pass

Table 8.1: Test Cases

8.3 Validation Process

The dataset was validated after preprocessing and before model training.

8.4 Performance Evaluation

Model performance was evaluated using classification metrics.

8.5 Limitations

- Dataset is limited to telecom customers.
- Real-time data was not available.

- External market factors were not included.

Chapter 9

Deployment

9.1 Deployment Process

The Power BI report was published to the Power BI Service.

9.2 Publishing and Sharing

Dashboards can be shared with stakeholders through secure links.

9.3 User Access and Usage

Role-based access can be configured for business users.

9.4 Documentation

User guides and dashboard documentation were prepared.

Chapter 10

Conclusion and Future Scope

10.1 Project Summary

The project successfully analyzed customer churn behavior using data analytics and machine learning techniques.

10.2 Objectives Achieved

All major objectives including preprocessing, EDA, dashboard development, and churn prediction were achieved.

10.3 Learning Outcomes

- Python programming
- Data preprocessing
- Exploratory data analysis
- Power BI dashboarding
- Machine learning with SVM

10.4 Future Enhancements

- Real-time churn prediction
- Deep learning integration
- Automated retention campaigns

- AI-based recommendation system
- Mobile dashboard application

References

1. Python Documentation
2. Pandas Documentation
3. NumPy Documentation
4. Matplotlib Documentation
5. Seaborn Documentation
6. Scikit-Learn Documentation
7. Power BI Documentation
8. Customer Analytics Research Papers

Appendix A

Dataset Samples

Sample records from the Telco Customer Churn dataset.

Appendix B

Dashboard Screenshots

Insert Dashboard 1, Dashboard 2, and Dashboard 3 screenshots.

Appendix C

Formulas and Calculations

Include churn rate, retention rate, and KPI formulas.

Appendix D

User Guide

Steps to use the Power BI dashboard and interpret KPIs.

Appendix E

Project Timeline

Include week-wise internship activities from Week 1 to Week 7.